

Question:

John plans to sell all his toys and use the money to buy video games. He has 13 lego sets and he sells them for \$15 each. He ends up buying 8 video games for \$20 each and has \$5 left. How many lego sets does he still have?

Reasoning + Answer:

<tool_call> Reasoning process here :

Step 1 : Calculate total money John earned from selling his Lego sets .
He has 13 Lego sets and sells each for \$ 15 .
So , total money earned = 13 × 15

Let's compute that :
 $13 \times 15 = (10 \times 15) + (3 \times 15) = 150 + 45 = 195$

So , John earned \$ 195 from selling the Legos .

Step 2 : Calculate total money he spent on video games .
He buys 8 video games at \$ 20 each .
Total spent = $8 \times 20 = 160$

Step 3 : He has \$ 5 left after buying the games .
So , total money he had = money spent + money left
Total money = $160 + 5 = 165$

But wait – we just said he earned \$ 195 from the Legos . That would mean he had \$ 195 , not \$ 165 .

There's a contradiction in our assumption – if he earned \$ 195 from 13 sets , and he only had \$ 165 left after spending , he can't have sold all his Lego sets and still had only \$ 5 .

Wait – the question says he "plans to sell all his toys" and uses the money to buy video games . Then says he "ends up buying 8 video games for \$ 20 each and has \$ 5 left" .

So the income from selling Legos must equal money spent on video games plus leftover money .

So :
Total money from Lego sales = money spent on games + leftover
 $= 8 \times 20 + 5 = 160 + 5 = 165$

So he sold Legos for \$ 165 .

He sells each Lego set for \$ 15 .

So total number of Lego sets sold = total money / price per set
 $= 165 / 15 = 11$

So John sold 11 Lego sets .

He originally had 13 sets .

So the number of sets he still has = $13 - 11 = 2$

Therefore , he still has 2 Lego sets .

<answer> 2 </answer>